LESSON 10
SPECIAL MULTIPLICATION

SUMMARY

10.1 Multiplication by 11
10.2 By One More than the One Before – a special type of multiplication.
10.3 Multiplication by Nines
10.4 The First by the First and the Last by the Last – a special type of multiplication.
10.5 Using the Average – of numbers to find their product.
10.6 Special Numbers – spotting factors of certain special numbers in a multiplication sum.

If there is an easy way to do a particular sum, rather than using the general method, we call it a special method. For example to multiply a number by 10 we do not use 'long multiplication'. In the Vedic system there are many special methods, which adds to the fun: the general method is always there but there is often a quick way if you can spot it.

The special methods play a large part in encouraging mental mathematics. Everyone likes a short cut, whether it is a quick way to get from one place to another or an easy way of doing a particular calculation. Life is full of special methods: to tackle all similar situations in the same way is not the way most people like to function. Every mathematical calculation invites its own unique method of solution and we should encourage children to look at the special properties of each problem in order to understand it best and decide on the best way forward. This is surely the intelligent way to do mathematics.

10.1 MULTIPLICATION BY 11

The 11 times table is easy to remember, and multiplying longer numbers by 11 is also easy.
If you want, say, 52×11 you want eleven 52's.
This means you want ten 52's and one 52 or 520 + 52:

$$\begin{array}{r} 5\,2\,0 \\ \underline{5\,2}\;+ \\ \underline{5\,7\,2} \end{array}$$

note how the 2 and the 5 get added in the middle column.

1 Find **52 × 11**.

To multiply a 2-figure number, like 52, by 11 you write down the number being multiplied, and put the total of the figures between the two figures: 5**7**2.

So **52 × 11 = 572**, between the 5 and 2 we put 7, which is 5+2.

✐ **Practice A** Multiply the following by 11:

a 23 × 11 **b** 61 × 11 **c** 44 × 11 **d** 50 × 11

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a 253 b 671 c 484 d 550

And so we can often quickly tell if a number can be divided exactly by 11.

2 Is **473** divisible by 11?

You can see that the middle number is the sum (total) of the outer numbers:
4 + 3 = 7.

So the number is divisible by 11.

In the example above you also know how many times 11 divides into 473.
It must be 43 because 43 × 11 = 473.

Just look at the outer numbers 4 and 3.

✐ **Practice B** Fill in the table below.

Number	Tick if Divisible	No. of Times it Divides
242		
594		
187		
791		
693		

Answers: 22, 54, 17, -, 63

"And as regards the time required by the students for mastering the whole course of Vedic Mathematics as applied to all its branches, we need merely state from our actual experience that 8 months (or 12 months) at an average rate of 2 or 3 hours per day should suffice for completing the whole course of mathematical studies on these Vedic lines instead of 15 or 20 years required according to the existing systems of Indian and also of foreign universities."
From "Vedic Mathematics", Page xvii.

CARRIES

Going back to multiplication by 11, there can sometimes be a carry, as the next example shows.

3 Find **58 × 11**.

The 5 and 8 here add up to 13 so the 1 has to be carried to the left:

58 × 11 = $5_1$38 = 638.

4 Find **47 × 11**.

The 4 and 7 here add up to 11 so again you carry 1 to the left:

47 × 11 = $4_1$17 = 517.

✎ **Practice C** Try these:

a 68 × 11 **b** 79 × 11 **c** 47 × 11

d 86 × 11 **e** 55 × 11 **f** 93 × 11

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a 748 b 869 c 517
d 946 e 605 f 1023

LONGER NUMBERS

This method can be easily extended to longer numbers.

5 Find **234 × 11**.

To multiply this 3-figure number by 11 you put the first and last figures of 234 as the first and last figures of the answer:

2 3 4

2 ? ? 4

Then for the second figure you add the first two figures of 234,
And for the third figure you add the last two figures of 234:

2 3 4

2 5 7 4

So **234 × 11 = 2574**.

Practice D Multiply the following by 11:

a 423 × 11 **b** 636 × 11 **c** 534 × 11

d 516 × 11 **e** 706 × 11 **f** 260 × 11

g 444 × 11 **h** 135 × 11 **i** 531 × 11

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a 4653 b 6996 c 5874
d 5676 e 7766 f 2860
g 4884 h 1485 i 5841

When you add the first or last two figures you could get a 2-figure number, so that there is a carry figure.

6 Find **777 × 11**.

The method above gives: $7_1 4_1 47$ = **8547**. We simply carry the 1's over, as before.

Practice E Multiply by 11:

a 384 × 11 **b** 629 × 11 **c** 888 × 11

d 555 × 11 **e** 393 × 11 **f** 939 × 11

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a 4224 b 6919 c 9768
d 6105 e 4323 f 10329

This can be extended to numbers of any size and also to multiplying by 111, 1111 etc.
This multiplication is useful in percentages work since if we want to increase a number by 10% we multiply it by 1.1, similarly with other percentage changes (see Manual 2 or The Cosmic Calculator, Book 2).

10.2 BY ONE MORE THAN THE ONE BEFORE

This special type of multiplication is for multiplying numbers whose first figures are the same and whose last figures add up to 10, 100 etc.

For example, 52 × 58, where both numbers start with 5 and 2 + 8 = 10.

7 Suppose we want to find **43 × 47** in which both numbers begin with 4 and the last figures (3 and 7) add up to 10.

Multiply 4 by the number *One More*: 4 × 5 = 20.

Then simply multiply the last figures together: 3 × 7 = 21.

So **43 × 47 = 2021** where 20 = 4×5, 21 = 3×7.

8 Similarly **62 × 68 = 4216** where 42 = 6×7, 16 = 2×8.

9 Find **204 × 206**.

Here both numbers start with 20, and 4 + 6 = 10, so the method applies.

204 × 206 = 42024 (420 = 20×21, 24 = 4×6).

✐ **Practice F** Multiply the following:

a 73 × 77	**b** 58 × 52	**c** 81 × 89	**d** 104 × 106
e 42 × 48	**f** 34 × 36	**g** 93 × 97	**h** 27 × 23
i 297 × 293	**j** 303 × 307		

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a 5621	**b 3016**	**c 7209**	**d 11024**
e 2016	**f 1224**	**g 9021**	**h 621**
i 87021	**j 93021**		

10 **93 × 39** may not look like it comes under this particular type of sum, but remembering the *Proportionately* formula we notice that 93 = 3×31, and 31 × 39 does come under this type:

31 × 39 = 1209 (we put 09 as we need double figures here)

so **93 × 39 = 3627** (multiply 1209 by 3)

The thing to notice in the last example is that the 39 needs a 31 for the method to work here: and then we spot that 93 is 3×31.

11 Finally, consider **397 × 303**.

Only the 3 at the beginning of each number is the same, but the rest of the numbers (97 and 03) add up to 100.

So again the method applies, but this time we must expect to have four figures on the right-hand side:

397 × 303 = 120291 where 12 = 3×4, 0291 = 97×3.

✐ **Practice G** Multiply the following:

a 64 × 38	**b** 88 × 46	**c** 33 × 74	**d** 66 × 28
e 36 × 78	**f** 46 × 54	**g** 298 × 202	**h** 391 × 309
i 795 × 705	**j** 401 × 499		

a 2432	**b 4048**	**c 2442**	**d 1848**
e 2808	**f 2484**	**g 60196**	**h 120819**
i 560475	**j 200099**		

10.3 MULTIPLICATION BY NINES

The Vedic formula *By One Less Than the One Before*, which is the converse of the formula *By One More than the One Before* comes in here in combination with *All From 9 and the Last From 10*.

12 **763 × 999 = 762/237**.

The number being multiplied by 9's is first reduced by 1: 763–1 = **762**. This is the first part of the answer.
Then *All From 9 and the Last From 10* is applied to 763 to get **237**, which is the second part of the answer.

13 **1867 × 99999 = 1866/98133**.

Here, as 1867 has 4 figures, and 99999 has 5 figures, we suppose 1867 to be 01867. This is reduced by 1 to give **1866** for the first part of the answer.
Then applying *All From 9. . .* to 01867 gives **98133** for the last part of the answer.

✐ **Practice H** Find the following:

a 89 × 99 **b** 82 × 99 **c** 19 × 99 **d** 45 × 99

e 778 × 999 **f** 7654 × 9999 **g** 79 × 999 **h** 124 × 9999

i 8989 × 99999 **j** 47 × 99999

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a 8811 b 8118 c 1881 d 4455
e 777222 f 76532346 g 78921 h 1239876
i 898891011 j 4699953

10.4 THE FIRST BY THE FIRST AND THE LAST BY THE LAST

Products like 43 × 47 are easy to find because the first figures are the same and the last figures sum to 10.

Similarly products like **27 × 87** are also easy to find because the **last figures are the same** and the **first figures add up to 10**.

This comes under the Vedic formula *The First by the First and the Last by the Last.*

14 **27 × 87 = 23/49**.

The conditions are satisfied here as 2 + 8 = 10
and both numbers end in 7.

So **we multiply the first figure of each number together and add the last figure**: 2 × 8 = 16, 16 + 7 = **23** which is the first part of the answer.

Multiplying the last figures together: 7×7 = **49**: which is the last part of the answer.

15 **69 × 49 = 3381**.

in which **33** = 6×4 + 9, and **81** = 9×9.

✐ **Practice I** Multiply the following by this method:

a 38 × 78 **b** 26 × 86 **c** 91 × 11 **d** 59 × 59

e 63 × 43 **f** 24 × 84 **g** 88 × 28 **h** 29 × 89

i 97 × 17 **j** 64 × 44

The following can also be done like this if you use the *Proportionately* formula as well:

k 31 × 42 **l** 46 × 83 **m** 93 × 71 **n** 88 × 32

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a 2964	**b 2236**	**c 1001**	**d 3481**
e 2709	**f 2016**	**g 2464**	**h 2581**
i 1649	**j 2816**		
k 1302	**l 3818**	**m 6603**	**n 2816**

10.5 USING THE AVERAGE

Here we look at a neat and easy way of multiplying numbers by using their average.
This comes under the formula *Specific General*.

16 Suppose we want to know **29 × 31**.
Since the average of 29 and 31 is 30, we might think that 29 × 31 is 30 × 30, or close to it.
In fact 29 × 31 = **899**
and this is just 1 below 900.

17 Now consider **28 × 32**. Again 30 is their average. 28 × 32 = **896** and this is 4 below 900.

18 For **27 × 33** whose average is also 30: 27 × 33 = **891**, which is 9 below 900.

In fact the rule is:

square the average and
subtract the square of the difference of either number from the average.

19 So **26 × 34** = $30^2 - 4^2$ = 900 – 16 = **884**.

And **58 × 62** = $60^2 - 2^2$ = 3600 – 4 = **3596**.

Similarly **94 × 106** = $100^2 - 6^2$ = 10,000 – 36 = **9964**.

And **37 × 33** = $35^2 - 2^2$ = 1225 – 4 = **1221**. See Section 12.1 for squaring numbers that end in 5.

This method is available for the product of any two numbers. Even if the average is not a very attractive number this method is still often better than multiplying the numbers. For example, for 67 × 69 it is easier to find $68^2 - 1$ than to multiply 67 by 69.

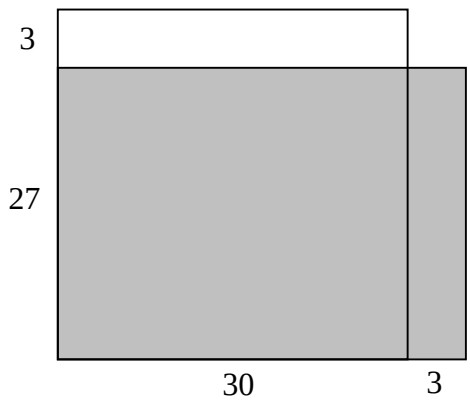
✐ **Practice J** Find:

a 49 × 51	**b** 27 × 33	**c** 57 × 63	**d** 64 × 66
e 85 × 65	**f** 55 × 95	**g** 33 × 47	**h** 91 × 99
i 44 × 48	**j** 43 × 47	**k** 74 × 86	**l** 98 × 102
m 62 × 38	**n** 48 × 72	**o** 73 × 93	**p** 196 × 204

a 2499	**b 891**	**c 3591**	**d 4224**
e 5525	**f 5225**	**g 1551**	**h 9009**
i 2112	**j 2021**	**k 6364**	**l 9996**
m 2356	**n 3456**	**o 6789**	**p 39984**

PROOF

A geometrical explanation for 27 × 33 is shown below.



The shaded rectangle is 27 by 33 and its area is 27 ×33.
The superimposed shape is a 30 by 30 square.
This shows that the square whose area is 30^2 is larger than the required rectangle by 3^2 units, as the top rectangle is 30 × 3 and the right-hand rectangle is 27 × 3, a difference of 3 ×3.

Here is an algebraic proof.

$\mathbf{(a + b)(a - b) = a^2 - b^2}$, where a is the average and b the difference of each number from the average. So (a + b) is the higher number and (a – b) is the lower number.

10.6 SPECIAL NUMBERS

REPEATING NUMBERS

Some multiplications are particularly easy.

20 **23 × 101 = 2323**.

To multiply 23 by 101 we need 23 hundreds and 23 ones, which gives 2323.

The effect of multiplying any 2-figure by 101 is simply to make it repeat itself.

21 Similarly **69 × 101 = 6969**.

And **473 × 1001 = 473473**.

Here we have a 3-figure number multiplied by 1001 which makes the 3-figure number repeat itself.

22 **47 × 1001 = 47047**.

Here, because we want to multiply by 1001, we can think of 47 as 047.
So we get 047047, or just 47047.

23 **123 × 101** = $123_1 23$ = **12423**.

Here we have 12300 + 123 so the 1 has to be carried over.

24 **28 × 10101 = 282828**.

✐ **Practice K** Find:

a 46 × 101 **b** 246 × 1001 **c** 321 × 1001

d 439 × 1001 **e** 3456 × 10001 **f** 53 × 10101

g 74 × 1001	**h** 73 × 101	**i** 29 × 1010101
j 277 × 101	**k** 521 × 101	**l** 616 × 101

a 4646	**b 246246**	**c 321321**
d 439439	**e 34563456**	**f 535353**
g 74074	**h 7373**	**i 29292929**
j 27977	**k 52621**	**l 62216**

This type of multiplication comes under the Sutra *By Mere Observation*.
Multiplications by 101 etc. are useful in percentages work as we multiply a number by 1.01 to increase it by 1% (see Manual 2 or The Cosmic Calculator, Book 2)

PROPORTIONATELY

25 **43 × 201 = 8643**.

Here we bring in the *Proportionately* formula: because we want to multiply by 201 rather than 101 we must put twice 43 (which is 86) then 43.

26 **31 × 10203 = 316293** we have 31×1, 31×2, 31×3.

✐ **Practice L** Find:

a 54 × 201	**b** 32 × 102	**c** 333 × 1003	**d** 41 × 10201	**e** 33 × 30201
f 17 × 20102	**g** 13 × 105	**h** 234 × 2001	**i** 234 × 1003	**j** 43 × 203

a 10854	**b 3264**	**c 333999**	**d 418241**	**e 996633**
f 341734	**g 1365**	**h 468234**	**i 234702**	**j 8729**

DISGUISES

Now it is possible for a sum to be of the above type without it being obvious: it may be disguised.

If we know the factors of some of these special numbers (like 1001, 203 etc.) we can make some sums very easy.

Suppose for example you know that **3 × 67 = 201**.

27 **93 × 67 = 6231**.

Since $3 \times 67 = 201$,
therefore $93 \times 67 = 31 \times (3 \times 67)$
$= 31 \times 201$
$= \mathbf{6231}$.

In other words, we recognise that one of the special numbers (201 in this case) is contained in the sum (as 3 × 67).

Now suppose we know that **3 × 37 = 111.**

28 **24 × 37 = 888**.

We know that $3 \times 37 = 111$, which is a number very easy to multiply.
So $24 \times 37 = 8 \times (3 \times 37)$
$= 8 \times 111$
$= \mathbf{888}$.

Also $19 \times 21 = 399 = 40\bar{1}$.

29 **38 × 63 = 2394**.

Since $38 \times 63 = 2 \times 19 \times 3 \times 21 = 6 \times (19 \times 21) = 6 \times 40\bar{1} = 240\bar{6} = \mathbf{2394}$.

If we know the factors of these special numbers we can make good use of them when they come up in a sum, and they arise quite frequently.

Below is a list of a few of these numbers with their factors:

67 × 3 = 201	**17 × 6 = 102**	**11 × 9 = $10\bar{1}$**
43 × 7 = 301	**13 × 8 = 104**	**19 × 21 = $40\bar{1}$**
7 × 11 × 13 = 1001	**29 × 7 = 203**	**23 × 13 = $30\bar{1}$**
3 × 37 = 111	**31 × 13 = 403**	**27 × 37 = $100\bar{1}$**

30 **62 × 39 = 2418**.

We see 31 × 13 contained in this sum:

$$\begin{aligned} 62 \times 39 &= 2\times31 \times 3\times13 \\ &= 2\times3 \times 31\times13 \\ &= 6 \times 403 \\ &= \mathbf{2418}. \end{aligned}$$

✐ **Practice M** Use the special numbers to find:

a 29 × 28	**b** 35 × 43	**c** 67 × 93	**d** 86 × 63
e 77 × 43	**f** 26 × 77	**g** 34 × 72	**h** 57 × 21
i 58 × 63	**j** 26 × 23	**k** 134 × 36	**l** 56 × 29
m 93 × 65	**n** 54 × 74	**o** 39 × 64	**p** 51 × 42

a 812	**b 1505**	**c 6231**	**d 5418**
e 3311	**f 2002**	**g 2448**	**h 1197**
i 3654	**j 598**	**k 4824**	**l 1624**
m 6045	**n 3996**	**o 2496**	**p 2142**

"These and many more interesting features there are in the Vedic decimal system, which can turn mathematics for the children from its present excruciatingly painful character to the exhilaratingly pleasant and even funny and delightful character it really bears ."

From "Vedic Mathematics", Page 239.